

MSSC 6250/Stat. Mach. Learn.

Instructor: Mehdi Maadooliat

Introduction



Department of Mathematical and Statistical Sciences



SYLLABUS - D2L - ONLINE MATERIALS

- [Syllabus](#)
- Course website: <http://tinyurl.com/SML-MU>

This is the homepage for MSSC 6250 - Statistical Machine Learning by [Dr. Mehdi Maadooliat](#) in Spring 2026 at Marquette University. All course materials will be posted on this site.

You can find the course syllabus [here](#) and the course schedule [here](#).

Class meetings

Meeting	Location	Time
Lecture	CU 208	Tue & Thur 12:30 pm - 1:45 pm
Office Hours	CU 351	Tue & Thur 1:45 - 3:15 pm

On this page
[Class meetings](#)

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Course information
Overview
Syllabus
Schedule
Useful links
Computing

Statistical Machine Learning

Dr. Mehdi Maadooliat
Marquette University
MSSC 6250 - Spring 2026

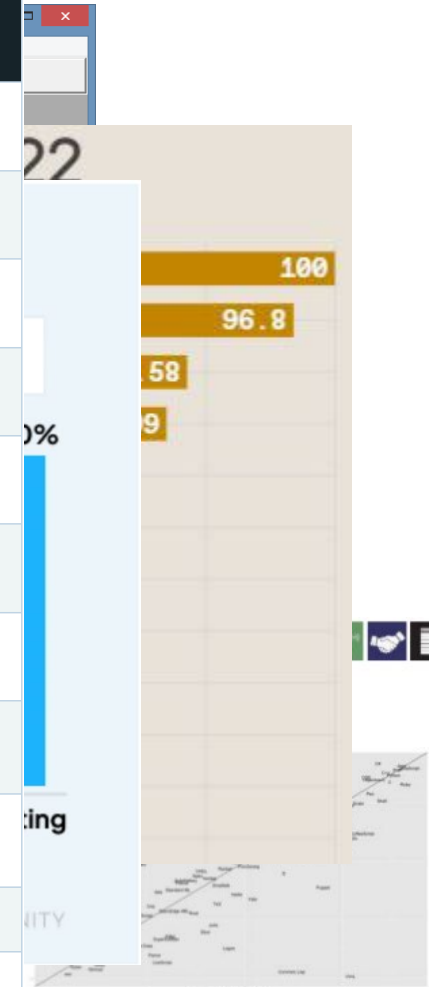
tinyurl.com/SML-MU

- Submissions (HW, Proj): <https://d2l.mu.edu/index.asp>

R VS PYTHON



Project Type	Recommended Language	Why
AI-Powered Applications	Python	Robust libraries like TensorFlow, PyTorch, and Scikit-learn simplify machine learning and AI development.
Statistical Analysis	R	Built-in statistical functions and libraries like Statsmodels and caret make R a top choice.
Web Applications	Python	Frameworks like Django and Flask enable efficient web app development and integration.
Big Data Processing	Python	Integrates with tools like Apache Spark and Dask for distributed data workflows
Academic Research	R	Focused on reproducibility, with strong support for hypothesis testing and data modeling
Real-Time Analytics	Python	Flexible for building pipelines with libraries like Pandas and integrating with streaming tools
Data Dashboards	R	Tools like Shiny and ggplot2 create interactive, publication-quality dashboards
Data Pipelines for E-commerce	Python	Handles large-scale data processing and real-time analytics for customer behavior
Healthcare Data Visualization	R	Creates detailed, professional-grade visualizations for healthcare analytics
Automation Scripts	Python	Libraries like Selenium and BeautifulSoup streamline automation tasks
Scientific Simulations	R	Offers statistical tools tailored for scientific modeling and research workflows
Interactive e-Learning Tools	Python	Supports web-based applications and AI-driven personalized learning systems



[RedMonk Programming Language Rankings, 2013](#)



HISTORY OF R (FROM [WIKIPEDIA](#))

- R is a [programming language](#) and software environment for [statistical computing](#) and graphics. The R language is widely used among [statisticians](#) and [data miners](#) for developing [statistical software](#) and data analysis. Polls, [surveys of data miners](#), and studies of scholarly literature databases show that R's popularity has increased substantially in recent years.
- R is an implementation of the [S programming language](#) combined with [lexical scoping](#) semantics inspired by [Scheme](#). [S](#) was created by [John Chambers](#) while at [Bell Labs](#). There are some important differences, but much of the code written for S runs unaltered.
- R was created by [Ross Ihaka](#) and [Robert Gentleman](#) at the [University of Auckland](#), New Zealand, and is currently developed by the *R Development Core Team*, of which Chambers is a member. R is named partly after the first names of the first two R authors and partly as a play on the name of [S](#).
- R is a [GNU project](#). The [source code](#) for the R software environment is written primarily in [C](#), [Fortran](#), and R. R is freely available under the [GNU General Public License](#), and pre-compiled binary versions are provided for various [operating systems](#). R uses a [command line interface](#); there are also several [graphical front-ends](#) for it.



R: THE STATISTICAL SOFTWARE

• Download and Install:

Windows

Comprehensive R Archive Network

Mac

The screenshot displays the RStudio environment. The R Console on the left shows the execution of R code, including data loading and plotting. The R Script Editor in the center contains R code for loading the 'ggplot2' package and creating a scatter plot of diamond prices. The R Package Manager on the right shows the installed packages, including 'graphics', 'grid', 'lattice', and 'methods'.

R Console Output:

```
> team2003$teamID
[1] ANA ARI ATL BAL BOS CHA CHN CIN CLE COL DET FLO HOU KCA LAN MIL MIN MON NYA
[20] NYN OAK PHI PIT SIN SEA SFN SLN TBA TEX TOR
148 Levels: ALT ANA ARI ATL BAL BFN BFF BL1 BL2 BL3 BL4 BLA BLF BLN BLU ... WSO
> team2003$payroll
[1] 79031667 80657000 106243667 73077500 99946500 51010000 79060313
[8] 59355667 40504834 67179667 49160000 49450000 71040000 40510000
[15] 105572620 40627000 55505000 51940500 152749014 116876429 50260813
[22] 70780000 54012429 45210000 86959167 82852167 83786666 19630000
[29] 103491667 51269000
> fitted(team2003$salary,lm)
1 2 3 4 5 6 7 8
82.58287 82.90759 86.01950 81.55313 86.76340 76.98448 82.75003 76.65184
9 10 11 12 13 14 15 16
76.49996 80.21498 76.61647 76.67281 80.98623 74.08803 87.08544 74.91008
17 18 19 20 21 22 23 24
77.08252 77.17198 97.31087 90.14380 76.03480 80.93429 77.74416 75.82571
25 26 27 28 29 30
84.16669 83.34616 83.53286 70.71513 87.46969 77.03622
> team2003$W
[1] 77 84 101 71 95 86 88 69 68 74 43 91 87 83 85 68 90
[20] 66 96 86 75 64 93 100 85 63 71 86
> pairs(team2003$forpairs)
> edit(team2003)
```

R Script Editor Code:

```
1 library(ggplot2)
2 source("plots/formatPlot.R")
3
4 view(diamonds)
5 summary(diamonds)
6
7 summary(diamonds$price)
8 aveSize <- round(mean(diamonds$carat), 4)
9 clarity <- levels(diamonds$clarity)
10
11 p <- qplot(carat, price,
12            data=diamonds, color=clarity,
13            xlab="Carat", ylab="Price",
14            main="Diamond Pricing")
15
```

R Package Manager:

status	Package	Description
loaded	graphics	The R Graphics Package
not loaded	grid	The Grid Graphics Package
not loaded	lattice	Lattice Graphics
loaded	methods	Formal Methods and Classes
not loaded	mon	CAME with CCV, cmath, base, animation

The R Graphics Package

documentation for package 'graphics' version 2.0.0

Help Pages

ABCDEFGHIJKLMNOPQRSTUVWXYZ

Diamond Pricing Plot:

The plot shows the relationship between Carat (x-axis) and Price (y-axis) for diamonds, colored by Clarity. The legend indicates the following clarity levels: I1, SI2, SI1, VS2, VS1, VVS2, VVS1, and IF.

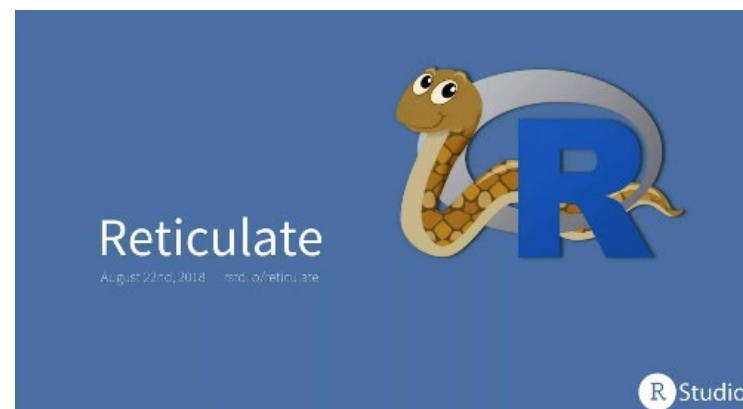
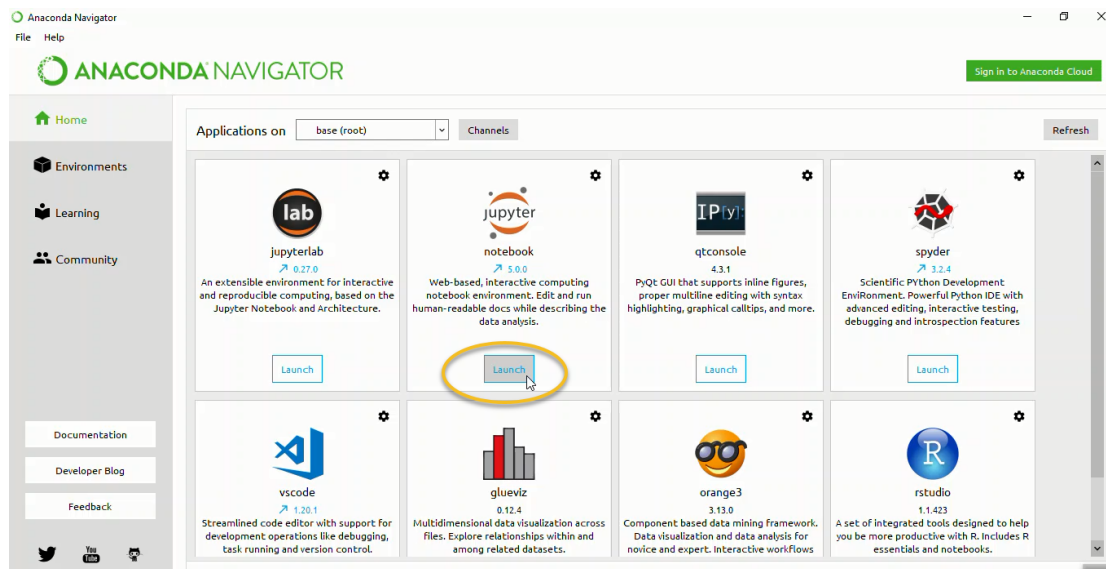
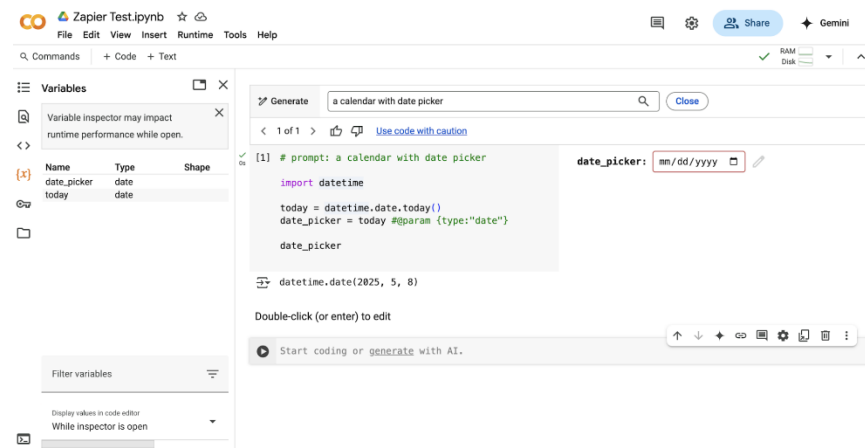
• RStudio:

- [Download](#)



PYTHON

(ANACONDA - JUPYTER - COLAB - RETICULATE):





MSSC 6250 CONT...

- Any General questions about Homework, Projects and Exams:

- **SHOULD** be posted in d2l Discussion Board.
- I will **NOT** answer general emails about Homework and Exams.

- Homework, Projects and Exams:

- Should be submitted as a **PDF** file (**Otherwise you will get ZERO**):
 - How to Combine Images into a PDF file [FREE & EASY + No Software] ([Youtube](#))
 - Microsoft Word to PDF in 10 Seconds ([Youtube](#))
 - How to: convert Images to PDF in Macbook/iMac ([Youtube](#))
 - <http://apple.stackexchange.com/questions/11163/how-do-i-combine-two-or-more-images-to-get-a-single-pdf-file>

MATH 1700 102 Mod Elementary Stat

Course Home | Content | Discussions | Dropbox | Quizzes | Classlist | Grades

Discussions List Subscriptions Group and Section Restrictions Statistics

New More Actions

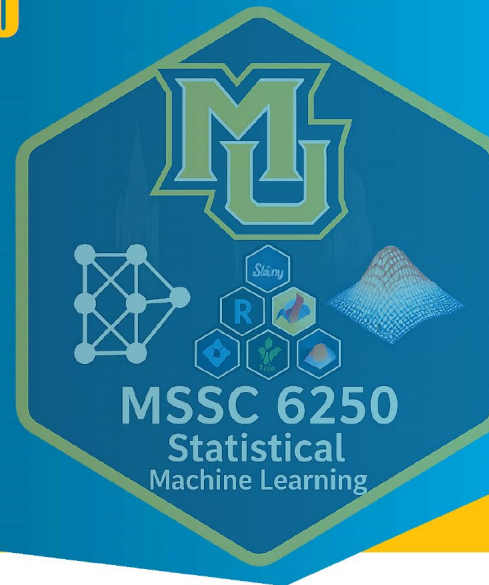
Filter by: Unread Unapproved

Homeworks

Hide Topics for Homeworks

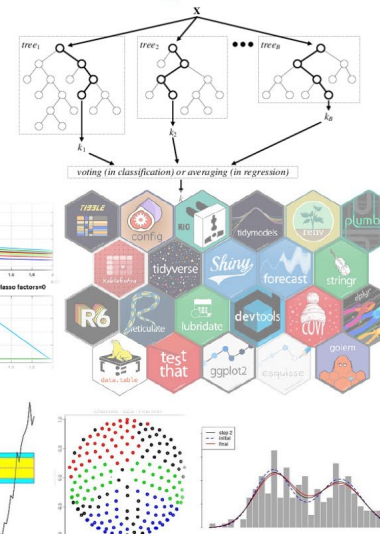
Topic	Threads	Posts
Homework 1	0	0
Homework 2	0	0
Homework 3	0	0

Statistical Machine Learning



Tentative topics :

- **Multivariate Data and Exploratory Analysis**
- **Multivariate Normal Distribution**
- **Supervised Learning:**
 - Classification and Regression
 - Logistic Regression and Linear Discriminant Analysis
 - Ridge Regression and Lasso
 - Tree Based Methods
 - Classification tree and Regression tree
 - Bagging, Random Forests, Boosting
 - Support Vector Machine
- **Unsupervised Learning** (Dim. Reduction and Clustering)
 - Principal Component Analysis (PCA)
 - Human genetic clustering
 - Google PageRank
 - Independent Component Analysis (ICA)
 - Blind Source Separation
 - K-Means and Hierarchical Clustering
- **Moving Beyond Linearity** (Smoothing Splines)
- **Natural Language Processing** (NLP)
 - Bag of Words; TF-IDF; Word2Vec
- **Neural Networks**
 - Introduction to Deep Learning
 - TensorFlow and Keras, or
 - PyTorch and fast.ai
 - Convolutional Neural Networks (CNN)
 - Recurrent Neural Networks (RNN)

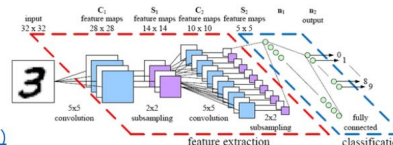


Prerequisites:

- A course in Statistical Methods
- A course in Linear Algebra

For more information, email the instructor:

- [Dr. Mehdi Maadooliat \(mehdi.maadooliat@marquette.edu\)](mailto:mehdi.maadooliat@marquette.edu)





SOME R RESOURCES

- [The R Project for Statistical Computing](#)
- [The Comprehensive R Archive Network](#)
- [R Reference Card 2.0](#)
- [MATLAB ® / R Reference](#)
- [An Introduction to R](#)
- [A \(Very\) Short Introduction to R](#)
- [R for Beginners](#)
- [simpleR - Using R for Introductory Statistics](#)
- [Using R for Data Analysis and Graphics](#)
- [Practical Regression and ANOVA using R](#)
- [R Markdown](#)
- [Quarto](#)



SOME PYTHON RESOURCES

- [Python Software Foundation](#)
- [Python Package Index \(PyPI\)](#)
- [NumPy Reference](#)
- [Pandas API Reference](#)
- [The Python Tutorial](#)
- [Learn Python in Y minutes](#)
- [Python for Data Analysis](#)
- [SciPy Lecture Notes](#)
- [Scikit-learn User Guide](#)
- [Jupyter Notebook](#)
- [Matplotlib](#)
- [Reticulate \(R-Python bridge\)](#)



REVIEW

- [Introduction to R and Programming](#)
- [Introduction to Python \(from ISLP\)](#)
- [Introduction to Linear Algebra](#)
- [Multivariate Normal Distribution](#)

QUESTIONS?

- **ANY QUESTION?**